

GIANIRA NICOLETTA ALFARANO

Universit de Rennes, France
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PERSONAL INFORMATION

Birthplace Noci (Bari), Italy
Birthdate 01/11/1993
Citizenship Italian
Webpage <https://user.math.uzh.ch/alfarano/>
ORCID <https://orcid.org/0000-0002-7463-6365>
ArXiv ID <https://arxiv.org/a/0000-0002-7463-6365>
Researchgate https://www.researchgate.net/profile/Gianira_Alfarano

RESEARCH INTERESTS

Algebraic coding theory, combinatorics, finite geometry, matroid theory, distributed storage, cryptography.

POSITIONS HELD

Chaire de Professeure Junior Universit de Rennes - IRMAR, France.	September 2024–current
Postdoctoral Researcher Vrije Universiteit Brussel (VUB), Belgium.	October 2023–2024
Postdoctoral Researcher Univrity College Dublin (UCD), Ireland.	October 2023–September 2024
Postdoctoral Researcher Eindhoven University of Technology (Tu/e), The Netherlands.	October 2022–September 2023
Teaching Assistant/Ph.D. student <i>University of Zurich</i> Advisor: Prof. Dr. Joachim Rosenthal	April 2018–September 2022
Research Internship Topic: Symmetric Cryptography HORST GÖRTZ INSTITUTE FOR IT-SECURITY (HGI), Ruhr Universität Bochum, Germany	February–June 2017

EDUCATION

University of Zurich, Switzerland Department of Mathematics Ph.D. student in Applied Algebra Thesis title: “Algebraic, Combinatorial and Geometric Aspects of Some Error-Correcting Codes” Adivsor: Prof. Dr. Joachim Rosenthal	2018–2022
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Università degli Studi di Trento, Italy Master of Mathematics - Curriculum Coding Theory and Cryptography Master degree in Mathematics <i>Grade: 110/110 cum Laude</i> Thesis title: “The Diffusion Property of some Mixing-Layers”	2015–2017
Università del Salento, Lecce, Italy Bachelor of Mathematics Bachelor degree in Mathematics <i>Grade: 110/110 cum Laude</i> Thesis title: “The classical unital”	2012–2015
Liceo Scientifico “Leonardo da Vinci”, Noci, Italy High school High school diploma <i>Grade: 100/100 cum Laude</i>	2007–2012

AWARDS AND GRANTS

Allocation d'installation scientifique (AIS) (10,000 EUR)	2027–2028
Dfis Scientifique (7,000 EUR)	2026
CreachLab–DGA , funding for a postdoc (140,000 EUR)	2026–2027
Program Bienvenue (Bretagne) , co-funding for a postdoc (85,000 EUR)	2027–2028
PHC Galilée , collaboration grant with University of Naples Federico II (8,000 EUR)	2026–2027
Basic Science (IBS), Daejeon, South Korea Invitation to participate to the workshop “2024 Workshop on (Mostly) Matroids” All expenses covered, including travel.	August 2024
FWO Junior Postdoctoral Fellowship Funded by Research Foundation - Flanders (FWO)	October 2024 – September 2027 amount: 200'000 EUR
Postdoc.Mobility Fellowship Funded by Swiss National Science Foundation Grant number 210966.	October 2022 – September 2024 amount: 106'600 CHF
The Applications of Computer Algebra Early Researcher Award 2022 https://math.unm.edu/ACA/ACA-ERA/Overview.html	August 2022 amount: 591 EUR
Best presentation award for the talk “The geometry of minimal codes”, given in the virtual summer school “Contemporary Algebraic and Geometric Techniques in Coding Theory and Cryptography”, Università degli Studi della Campania, Luigi Vanvitelli.	July 2022 amount: 500 EUR
UZH Alumni grant Additional grant for the organization of the Summer School “ACT 22: Algebraic Coding Theory”	February 2022 amount: 2'500 CHF
Ph.D Mobility Fellowship Financial support from the Swiss National Science Foundation for 6-months scientific visit	February–August 2021

University College Dublin, Ireland.

amount **12'000 CHF**

Grant GRC 2019-Q3-G-003

October 2020

Grant for the organization of the Summer School

“ACT 22: Algebraic Coding Theory”

amount: **9'000 CHF**

VAUZ

2020

Additional grant for the organization of the Summer School

“ACT 21: Algebraic Coding Theory”

amount **1'000 CHF**

Financed through the Swiss National Science Foundation

grant no. 188430

September 2019 – June 2022

P.I. Prof. J. Rosenthal

Oberwolfach Leibniz Graduate Student (OWLG)

March 2019

grant to participate to the

Oberwolfach Workshop 1912, Contemporary Coding Theory

Oberwolfach, Germany.

Schloss Dagstuhl - Leibniz Center for Informatics

December 2018

Invitation to participate to the seminar

“Algebraic Coding Theory for Networks, Storage, and Security” (18511), Germany.

OTHER AWARDS

Travel Grant (registration fee and accommodation)

19 February- 25 February 2023

Financial support for “Alcocrypt 2023” from the local committee, CIRM, Marseille, France.

Travel Grant (registration fee)

15 August- 19 August 2022

Partial financial support for “ACA 2022” from the local committee, in Istanbul, Turkey.

Travel Grant: 500 euros

26 June- 1 July 2022

Partial financial support for “ISIT 2022” from the local committee, in Espoo, Finland.

Travel Grant (registration fee)

30 May- 3 June 2022

Partial financial support for “Combinatorics 2022” from the local committee, in Mantua, Italy.

ACADEMIC VISITS

Invited researcher at VUB Brussels University, Belgium, 17-22 January 2027.

Invited researcher at University of Zurich, Switzerland, 22-31 May 2025.

Invited researcher at University of Naples, Italy, 28 October-01 November 2024.

Invited researcher at University of Bielefeld, Germany, 26-30 May 2024.

Invited researcher at University of Zurich, Switzerland, 20-31 August 2023.

Invited researcher at University of Zurich, Switzerland, 6-9 December, 2022.

Invited researcher at University of Paris VIII, France, November 23-December 5, 2022.

Invited researcher at University of Paris VIII, France, 5-18 September 2021.

Invited researcher at the Max Planck Institute for Mathematics in the Sciences (MiS) in Leipzig, Germany, July-August 2021.

Researcher at University College Dublin, Ireland, February–August 2021

Invited researcher at the Max Planck Institute for Mathematics in the Sciences (MiS) in Leipzig, Germany, 21-23 September 2020.

Invited researcher at the Universidad de Alicante, Alicante, Spain, May-June 2019.

Research internship at HORST GÖRTZ INSTITUTE FOR IT-SECURITY (HGI), Ruhr Universität, Bochum, Germany, February-June 2017

TITLES

Italian **qualification as associate professor** (II fascia) in “Geometry and Algebra”. 2024– 2036

PUBLICATIONS IN PEER-REVIEWED JOURNALS

23. **G. N. Alfarano**, S. Degen. The polytope of all q -rank functions. *To appear in SIAM Journal on Discrete Mathematics*, **2026**.
arXiv version, <https://arxiv.org/abs/2505.08018>
22. **G. N. Alfarano**, E. Byrne. Recursive properties of the characteristic polynomial of weighted lattices. *Advances in Applied Mathematics*, **2026**.
21. **G. N. Alfarano**, E. Byrne, A. Fulcher. The Free Product of q -Matroids. *The Electronic Journal of Combinatorics*, **2025**.
20. **G. N. Alfarano**, R. Jurrius, A. Neri, F. Zullo, Representability of the direct sum of uniform q -matroids. *To appear in Combinatorial Theory*, **2025**.
arXiv version, <https://arxiv.org/abs/2408.00630>
19. A. Abiad, **G. N. Alfarano**, A. Ravagnani. Eigenvalue bounds and alternating rank-metric codes. *Journal of Algebra and its Applications*, **2025**.
18. **G. N. Alfarano**, J. Rosenthal, B. Toesca. Schubert Subspace Codes. *Journal of Algebra and its Applications*, **2025**.
17. **G. N. Alfarano**, K. Khathuria, and S. Tinani. On Cyclic Matroids and their Applications. *The Bulletin of the Institute of Combinatorics and its Applications*, Vol. 104 **2024**.
16. **G. N. Alfarano**, A. Neri, F. Zullo. Maximum Flag-Rank Distance Codes. *Journal of Combinatorial Theory, Series A*, **207 (2024)**: 105908.
DOI: 10.1016/j.jcta.2024.105908
15. **G. N. Alfarano**, A. B. Kılıç, A. Ravagnani, E. Soljanin. The Service Rate Region Polytope. *SIAM Journal on Applied Algebraic Geometry*, Vol. 8, Iss. 3 **2024**
DOI: 10.1137/23M1557829
14. **G. N. Alfarano**, E. Byrne. The Cyclic Flats of a q -Matroid. *Journal of Algebraic Combinatorics*, **2024**.
DOI:10.1007/s10801-024-01321-2
13. **G. N. Alfarano**, M. Borello, A. Neri. Outer Strong Blocking Sets. *The Electronic Journal of Combinatorics*, **2024**.
DOI:10.37236/12046
12. **G. N. Alfarano**, A. B. Kılıç, A. Ravagnani. Service Aspects of LRC and Batch Codes. *IEEE BITS the Information Theory Magazine*, **2024**.

DOI:10.1109/MBITS.2023.3345806

11. **G. N. Alfarano**, D. Napp, A. Neri, and V. Requena. Weighted Reed-Solomon Convolutional Codes. *Linear and Multilinear Algebra*, **2022**.
DOI: 10.1080/03081087.2023.2169232
10. **G. N. Alfarano**, M. Borello, A. Neri, and A. Ravagnani. Linear Cutting Blocking Sets and Minimal Codes in the Rank Metric. *Journal of Combinatorial Theory, Series A*, **2022**.
DOI: 10.1016/j.jcta.2022.105658
9. **G. N. Alfarano**, A. Gruica, J. Lieb, J. Rosenthal. Convolutional codes over finite chain rings, MDP codes and their characterization. *Advances in Mathematics of Communications*, **2022**.
DOI: 10.3934/amc.2022028
8. **G. N. Alfarano**, F.J. Lobillo, A. Neri, A. Wachter-Zeh. Sum-rank product codes and bounds on the minimum distance. *Finite Fields and their Applications*, **2022**.
DOI: 10.1016/j.ffa.2022.102013
7. **G. N. Alfarano**, M. Borello, A. Neri, and A. Ravagnani. Three Combinatorial Perspectives on Minimal Codes. *Siam Journal on Discrete Mathematics*, **2021**.
DOI: 10.1137/21M1391493
6. **G. N. Alfarano**, J. Lieb, J. Rosenthal. Construction of LDPC convolutional codes via difference triangle sets. *Designs, Codes and Cryptography*, **2021**.
DOI: 10.1007/s10623-021-00912-5
5. **G. N. Alfarano**, K. Khathuria, V. Weger. A survey on single server private information retrieval in a coding theory perspective. *Applicable algebra in engineering communication and computing*, **2021**.
DOI: 10.1007/s00200-021-00508-5
4. **G. N. Alfarano**, F.J. Lobillo, and A. Neri. Roos bound for skew cyclic codes in Hamming and rank metric. *Finite Fields and Their Applications*, 69: 101772, **2021**.
DOI: 10.1016/j.ffa.2020.101772
3. **G. N. Alfarano** and J. Lieb. On the left primeness of some polynomial matrices with applications to convolutional codes. *Journal of Algebra and its applications*, **2020**.
DOI: 10.1142/S0219498821502078
2. **G. N. Alfarano**, M. Borello, and A. Neri. A geometric characterization of minimal codes and their asymptotic performance. *Advances in Mathematics of Communications*, **2020**.
DOI: 10.3934/amc.2020104
1. **G. N. Alfarano**, C. Beierle, T. Isobe, S. Kölbl, G. Leander. ShiftRows Alternatives for AES-like Ciphers and Optimal Cell Permutations for Midori and Skinny. *IACR Transactions of Symmetric Cryptology*. 2018(2): 20-47, **2018**.
DOI: 10.13154/tosc.v2018.i2.20-47

REFEREED CONFERENCE PROCEEDINGS

3. **G. N. Alfarano**, M. Le Meur. Minimal Additive Codes. WCC 2026.
2. **G. N. Alfarano**, A. Ravagnani, E. Soljanin. Dual-Code Bounds on Multiple Concurrent (Local) Data Recovery. *2022 IEEE International Symposium on Information Theory (ISIT)*, **2022**.
DOI: 10.1109/ISIT50566.2022.9834616.
1. **G. N. Alfarano**, J. Lieb, and J. Rosenthal. Construction of Rate $(n - 1)/n$ Non-Binary LDPC Convolutional Codes via Difference Triangle Sets. *2020 IEEE International Symposium on Information Theory (ISIT)*, **2020**.

PREPRINTS

6. **G. N. Alfarano**, A. Neri, B. Toesca. Combinatorial constructions of Schubert subspace codes. *submitted*. arXiv version, <https://arxiv.org/abs/2607.07479>
5. **G. N. Alfarano**, M. Le Meur. Minimal additive codes and additive strong blocking sets. *submitted*. arXiv version, <https://arxiv.org/abs/2606.24262>
4. **G. N. Alfarano**, G. Marino, A. Neri, R. Trombetti. A geometric approach to generalized covering radii of linear codes. *submitted*. arXiv version, <https://arxiv.org/abs/2606.16669>
3. **G. N. Alfarano**, U. Mushrraf, F. Zullo. On perfect flag-rank metric codes. *submitted*. arXiv version, <https://arxiv.org/abs/2606.12934>
2. **G. N. Alfarano**, M. Borello, A. Neri. The geometry of rank-metric codes. *submitted*. arXiv version, <https://arxiv.org/abs/2605.19691>
1. **G. N. Alfarano**, S. Degen. Representability of q -matroids via rank-metric codes. *submitted*. arXiv version, <https://arxiv.org/abs/2605.15780>

INVITED TALKS

Plenary talks in conferences are marked with (*)

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| 39. A geometric approach to linear codes with applications to one-weight codes
Online seminar, UMI Crittografia e Codici group. | May 2026 |
| 38. A geometric approach to linear codes with applications to one-weight codes
Seminar in Grenoble, organized by the group CASC.
Grenoble, France. | April 2026 |
| 37. (*) Skew-polynomial rings and algebraic coding theory
Groebner free methods and their applications.
Barcelona, Spain. | April 2026 |
| 36. (*) The geometry of one-weight codes
Journées Codage et Cryptographie 2026
Cte d'Opale, France. | March 2026 |
| 35. (*) Skew-polynomial rings and algebraic coding theory
Applications of Computer Algebra (ACA25).
Heraklion (Crete), Greece. | July 2025 |
| 34. Maximum Flag Rank Distance Codes
SIAM Conference on Applied Algebraic Geometry (AG25).
Special Session: Network Coding
University of Wisconsin, USA. | July 2025 |
| 33. The geometry of one-weight codes
Dutch-Belgian discrete math seminar
Bruxelles, Belgium. | March 2025 |
| 32. The geometry of one-weight linear codes
Zurich Seminar in Coding Theory and Cryptography. | May 2025 |

31. **The service rate region polytope** May 2025
Seminar Discrete Mathematics, Codes and Cryptography - University of Paris 8.
30. **The geometry of one-weight linear codes** May 2025
London Symposium on Information Theory,
Cambridge, UK.
29. **Schubert Subspace Codes** Mars 2025
Algebraic Applications to Coding Theory and Crypto
Mexico City.
28. **Minimal Codes: The Role of Supports in Coding Theory** January 2025
Barracuda meeting (ANR)
Bayeux, France.
27. **Constant Dimension Subspace Codes in Schubert Varieties** July 2024
2nd AMS-UMI International Joint Meeting.
Special Session: Algebraic Coding Theory.
Palermo, Italy.
26. **Representability of q -matroid products and linear sets** May 2024
DOCTOR: Delightful Occasion for Combinatorial Talks on Outstanding Research
Vrije Universiteit Brussel
25. **Representability of q -matroid products and linear sets** May 2024
Research Seminar
Bielefeld University
24. (*)**An Overview on Algebraic Coding Theory** May 2024
Workshop Women in Mathematics
Ghent University
23. **An Overview on Minimal Codes and Strong Blocking Sets** March 2024
ACCESS Seminars (online)
22. (*)**The Critical Theorem for Rank-Metric Codes** February 2024
OpeRa 2024 - Caserta Open Problems on Rank-Metric Codes.
Università degli studi della Campania, Luigi Vanvitelli.
21. **The Geometry of Minimal Codes** January 2024
Gomtrie et algbre effectives.
Institut de recherche mathmatique de Rennes, France.
20. **Rank-metric codes, q -matroids, and independent spaces** December 2023
International Conference on Algebraic Geometry, Coding Theory and Combinatorics - In honour
of Sudhir Ghorpades 60th birthday
IIT Hyderabad (India)
19. (*)**The Service Rate Region Polytope** November 2023
Colloquium on Coding Theory.
Ghent University, Belgium.
18. **The Geometry of Minimal Codes** October 2023
Algebra and Number Theory Seminars.
University College Dublin, Ireland.
17. (*)**The Geometry of Minimal Codes** September 2023
Annual Conference on Cryptography and Coding Theory - gruppo UMI.
University of Perugia Italy.

16. **Minimal Codes and Strong Blocking Sets** July 2023
 SIAM Conference on Applied Algebraic Geometry (AG23).
 Special Session: Coding Theory and Galois Geometries.
 Eindhoven, The Netherlands.
15. **Maximum Flag Rank Distance Codes** July 2023
 29th Nordic Congress of Mathematicians with EMS.
 Special Session: Algebraic Coding Theory and Cryptography.
 Aalborg, Denmark.
14. **Rank-Metric Codes and q -Matroids** April 2023
 University of Groningen, The Netherlands.
13. **Matroidal Approach to Coding Theory** December 2022
 University of Zurich, Switzerland.
12. **Critical Problem for q -Polymatroids** December 2022
 University of Paris VIII, France.
11. **Critical Problem, q -Polymatroids and Rank-Metric Codes** August 2022
 ACA 2022, Session “Algebraic and Geometric Methods in Coding Theory”, Istanbul, Turkey.
10. **q -Analogues of matroids and codes** Apr 2022
 Max Planck Institute for Mathematics in the Sciences, Leipzig, Germany.
9. **(Linear) cutting blocking sets and minimality of codes** October 2021
 Sabancı Algebra Seminar, Turkey
8. **(Linear) cutting blocking sets and their codes** September 2021
 PICS, Postgraduate International Coding theory Seminar.
7. **Bounds on the distance of skew cyclic codes** September 2021
 Seminar Discrete Mathematics, Codes and Cryptography - University of Paris 8.
6. **A geometric interpretation of minimal code** May 2021
 De Cifris, Schola Latina Seminars.
5. **Minimal linear codes and cutting blocking sets** March 2021
 Maths Postgraduate Seminar Series, University College Dublin.
4. **The geometry of minimal linear codes** March 2021
 eSeminar UGent-VUB.
3. **What is...a convolutional code?** December 2019
 Zurich Graduate Colloquium, Zurich, Switzerland.
2. **σ -Quasicyclic and Projective Convolutional Codes** June 2019
 University of Alicante, Alicante, Spain.
1. **An introduction to Convolutional Codes** June 2018
 Research Seminars, University of Trento, Trento, Italy.

OTHER TALKS

13. **Minimal additive codes.** May 2026
 Combinatorics 2026
 Naples, Italy.

12. **Direct sum of q -matroids and linear sets.** August 2024
Workshop on Mostly Matroids 2024, Institute for Basic Science (IBS).
Daejeon, South Korea.
11. **Free product of q -matroids and clubs on the projective line.** June 2024
Combinatorics 2024, Carovigno, Italy.
10. **Minimal Codes and Strong Blocking Sets** July 2023
International Conference on Finite Fields and Their Applications 2023 (Fq15).
Paris, France.
9. **Weighted Reed-Solomon Convolutional Codes** June 2023
Workshop on Convolutional Codes.
Zurich, Switzerland.
8. **Cyclic Spaces, q -Matroids and Rank-Metric Codes** January 2023
Dutch-Belgian discrete math seminar
Ghent University, Ghent (Belgium).
7. **q -Matroids and rank-metric codes** August 2022
ACA 2022, Session “ q -analogues in combinatorics: matroids, designs and codes”, Istanbul, Turkey.
6. **The geometry of minimal codes** July 2022
“Contemporary Algebraic and Geometric Techniques in Coding Theory and Cryptography”,
(virtual summer school) Università degli Studi della Campania, Luigi Vanvitelli.
5. **Dual-Code Bounds on Multiple Concurrent (Local) Data Recovery** June 2022
IEEE International Symposium on Information Theory (ISIT), Aalto University, Espoo, Finland.
4. **q -Matroids and rank-metric codes** June 2022
Combinatorics 2022, Mantua, Italy.
3. **Roos-like bound for skew-cyclic codes in Hamming and (sum-)rank metric** September 2021
First Annual Conference: Cryptography and Coding Theory.
2. **Construction of Rate $(n-1)/n$ Non-Binary LDPC Convolutional Codes via Difference Triangle Sets** June 2020
IEEE International Symposium on Information Theory (ISIT), Los Angeles, California, (virtual conference).
1. **New Algebraic Construction of MDP Convolutional Codes** December 2019
Arbeitsgemeinschaft in Codierungstheorie und Kryptographie, University of Neuchatel, Switzerland.

OTHER ACTIVITIES

Lecturer at the NonCommutative Rings and their Applications, IX

“Skew-Polynomial Rings and Linearized Polynomials in Algebraic Coding Theory”, 4 hours. July 2025
Lens, France.

Lecturer at the CIMPA School Finite Geometry and Coding Theory

for the course “Algebraic Curves over Finite Fields”, 4 hours.
Sangareddy, India.

November 2023

Tutor at the CIMPA School Finite Geometry and Coding Theory

for the courses “Algebraic Curves over Finite Fields” (2 hours) and

“Lattices and Codes” (2 hours)
Sangareddy, India.

November 2023

“Matematica per la crittografia”

Laboratory of Cryptography for High School Students, 6 hours, in italian.
Liceo Scientifico, Leonardo da Vinci, Noci, Italy.

May-June 2021

TEACHING EXPERIENCE

Lecturer and Organizer

Algorithmique de base
Université de Rennes, France

September 2024–current

Invited Lecturer

Algebra and Geometry of Rank-Metric Codes
Doctoral Course in Mathematics (10 hours)
University of Naples Federico II, Italy

January 2026

Invited Lecturer

Strong blocking sets and their applications in coding theory
Doctoral Course in Mathematics
University of Naples Federico II, Italy

January 2025

Lecturer and Organizer

MAT011 Matroids theory
University of Zurich, Switzerland

February 2022–June 2022

Teaching Assistant

University of Zurich, Switzerland
Duties: frontal lectures, exercise classes,
preparation and correction of exams and homeworks.

April 2018–August 2022

MAT476 Finite Fields, Theory and Applications (Fall 2020)

MAT512 Elliptic curves (Spring 2018, Spring 2020)

MAT001 Kryptographie (Fall 2019)

MAT005 Codierungstheorie (Spring 2019)

MAT211 Algebra (Fall 2018)

Tutor and Teaching Assistant

University of Trento, Italy
Duties: frontal lectures, exercise classes.
Analisi Matematica I

March 2015–December 2017

MENTORING EXPERIENCE

• **Ph.D. Students**

- Marine Le Meur
Université de Rennes, France
jointly with Delphine Boucher

September 2025 - current

• **Postdocs**

- Rakhi Pratihar
Université de Rennes, France

November 2025 - current

- Adrien Vincotte January 2026 - current
 Université de Rennes, France
- **Bachelor and Master Students**
- L1 stage d’initiation la recherche: Jannys Bellevergue June 2026
 Université de Rennes, France
 Project title: “Sphere packing bound and perfect codes”
- L1 stage d’initiation la recherche: Edan Poulain June 2026
 Université de Rennes, France
 Project title: “Sphere packing bound and perfect codes”
- L3 stage: Pierre Badra May 2025 - June 2025
 Université de Rennes, France
 Project title: “Bounds on cyclic Codes”
- M2 stage: Marine Le Meur April 2025 - July 2025
 Université de Rennes, France
 Project title: “The Geometry of Additive Codes”
- M2 stage: Aaron Bundi Anampiu November 2024 - January 2025
 Université de Rennes, France
 Project title: “On Zeros of a Polynomial in a Finite Grid”
- Master student co-supervised: B. Toesca di Castellazzo May 2022 - June 2023
 University of Zurich, Switzerland
 jointly with Prof. Dr. J. Rosenthal
 Thesis title: “Maximal Partial Spreads in Schubert Varieties”
- Master student co-supervised: N. Zürcher September 2019 - August 2020
 University of Zurich, Switzerland
 jointly with Prof. Dr. J. Rosenthal
 Thesis title: “On the Intractability of Maximum-Likelihood Decoding
 for Reed–Solomon Self-Dual Codes”
- Master student co-supervised: A. Gruica August 2019 - April 2020
 University of Zurich, Switzerland
 jointly with Prof. Dr. J. Rosenthal
 Thesis title: “MDP Convolutional Codes over $\mathbb{Z}/p^r\mathbb{Z}$ ”

MEMBERSHIPS IN BOARDS AND INDIVIDUAL REVIEWING ACTIVITIES

Evaluator for PhD dissertations:

- Sam Adriaensen, Vrije Universiteit Brussel, Belgium May 2024
- Jonathan Mannaert, Vrije Universiteit Brussel, Belgium June 2025
- **Examinator** for Martin Scotti, University of Paris 8, France October 2025
- **Examinator** for Matteo Abbondati, University of Montpellier, France November 2025
- **Examinator** for Charles Brion, University of Rouen, France October 2026

Program Committee: WCC 2026

Reviewer for the following scientific journals: Sep 2018 – current
 “Advances in Mathematics of Communication”, “IEEE Transactions on Information Theory”, “Finite Fields and their Applications”, “Designs Codes and Cryptography”, “Discrete Mathematics”, “Journal of Algebra and its Applications”, “Cryptography and Communications”, “IEEE Communication Letters”, “The Ramanujan Journal”, “The Electronic Journal of Combinatorics”, “Mathematische Annalen”, “Bulletin of the London Mathematical Society”.

Reviewer for the following conferences:
“IEEE ISIT”, “IEEE ISTC”, “WCC”.

Sep 2019 – current

ORGANIZATIONAL AND PROFESSIONAL EXPERIENCE

Co-organizer of the Conference “32th Applications of Computer Algebra (ACA 2027)” July 5–9, 2027

Rennes, France.

Link: <https://aca2027.sciencesconf.org/>

Co-organizer of the Workshop “Open Problems on Rank-Metric Codes 2026” February 23–27, 2026
Bordeaux, France.

Link: <https://www.math.univ-paris13.fr/~borello/OpeRa2026/main>

Co-organizer of the Session “Combinatorial and Geometrical Methods in Contemporary Coding Theory” for the conference “30th Applications of Computer Algebra (ACA 2025)” July 14–18, 2025
Heraklion, Greece.

Link: <https://www.giovannilongobardi.it/aca2025>

Co-organizer of the Session “Algebraic and Geometric Methods in Coding Theory (AGMCT)” for the conference “3SIAM Conference on Applied Algebraic Geometry - AG25” July 7–11, 2025
Heraklion, Greece.

Link: <https://www.math.univ-paris13.fr/~borello/SIAMAG25-AGMCT>

Co-organizer of “Workshop on Convolutional Codes” June 5 – 9, 2023
University of Zurich, Switzerland.

Link: <https://www.math.uzh.ch/convolutionalcodes>

Co-organizer of the Session “Network Coding” for the conference “25th International Symposium on Mathematical Theory of Networks and Systems (MTNS 2022)” September 12–16, 2022
Bayreuth, Germany.

Link: <https://www.mtns2022.uni-bayreuth.de/en/index.html>

Co-organizer of the Session “ q -analogues in combinatorics” for the conference “ACA 2022 - Applications of Computer Algebra” August 15 – 19, 2022
Istanbul-Gebze, Turkey

Link: <https://sites.google.com/view/michela-ceria-home-page/aca-2022-q-mat>

Co-organizer of the events “Female in math get-togethers” October 2021–August 2022
University of Zurich and ETH Zurich

Co-organizer of the seminars “Zurich Graduate Colloquium” February 2021–August 2022
Zurich Graduate School in Mathematics

Co-organizer of “ACT 21: Algebraic Coding Theory e-Summer School” June 7 – 11, 2021
University of Zurich, Switzerland

Link: <https://www.math.uzh.ch/act/>

Co-organizer of “ACT 22: Algebraic Coding Theory Summer School” July 4 – 8, 2022
University of Zurich, Switzerland

Link: <https://www.math.uzh.ch/act/>

Co-organizer of “Coding Theory and Cryptography,

A conference in honor of
Joachim Rosenthal's 60th birthday”
University of Zurich, Switzerland
Link: <https://www.math.uzh.ch/aa/index.php?id=32>

July 11 – 15 , 2022

MEMBERSHIPS

Member of De Componendis Cifris	January 2024 – current
Member of the Unione Matematica Italiana (UMI)	Jan 2024 – current
Member of European Women in Mathematics	Jan 2024 – current
Member of the European Mathematical Society (EMS)	June 2023 – current